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CS5002 Senior Design II

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Final Self-Assessment

Throughout the senior design project, my primary contributions focused on integrating external APIs into our system and implementing user authentication across both the frontend and backend. I worked extensively on connecting sports and esports APIs to our backend so that we could retrieve and store match results in our database. This required handling differences in data formats across leagues and ensuring that the data could be processed consistently within our system. In addition, I fully implemented user authentication, including login and registration functionality, and ensured that it worked seamlessly across the backend and frontend. This process involved managing user credentials securely, validating login attempts, and connecting the authentication flows to our database. I also assisted with testing and debugging to make sure that these systems integrated properly with the rest of the application. Overall, my role centered on making sure that real-world data and user accounts functioned reliably within Bracketed.

This work built directly upon the skills I identified in my initial self-assessment, particularly in software engineering, data management, and system design. My experience in Database Design and Development helped me understand how to structure and store API data efficiently, while my background in Software Engineering guided my approach to organizing backend logic and working within a team environment. One of my biggest

successes was successfully integrating external APIs and ensuring that match data could be stored and used within our application, despite inconsistencies between different leagues. I also gained valuable experience implementing authentication, which strengthened my understanding of secure user management. A major challenge that I faced was handling the various API data formats and ensuring that they could be made uniform for use across the platform. Working through these challenges improved my problem-solving skills and taught me how to design more flexible systems that can be adapted for various data sources. Overall, I expanded both my technical skills and my confidence in contributing to a full-scale software project.

As a group, we successfully developed a functional prototype of Bracketed that integrates a frontend interface, backend services, a database, and external APIs. We implemented key features such as user authentication, match data retrieval, prediction tracking, and challenge-based competition between users. One of our biggest accomplishments was connecting all major components of the system so that data flows consistently from external APIs into our database and is then displayed on the frontend. Throughout the project, we were able to divide responsibilities based on each member's strengths or areas that they wanted to improve, allowing us to make steady progress across multiple areas of development. Overall, our group achieved the goal of building a cohesive and working application that reflects a full-stack development effort.

In terms of teamwork, collaboration and communication were key factors in our success. We regularly shared updates, helped each other troubleshoot issues, and held weekly meetings to ensure consistent progress. One challenge we faced was balancing the

project with other academic responsibilities, which sometimes made it difficult to maintain consistent progress across all areas at the same time. Despite this challenge, each team member contributed meaningfully, and the workload was distributed fairly evenly based on our assigned roles. Sam focused mainly on the frontend, Luke focused on the integration between the frontend and backend, I focused on the backend, and Michael focused on the database. My contributions aligned well with the rest of the team, particularly in API integration and authentication while others focused on frontend development and database design. While the assigned workload was fairly even, Luke deserves some special recognition for picking up extra work when others were falling behind due to other responsibilities, especially on the database. He implemented NeonDB, allowing us to quickly get back on track for our database progress. Aside from this special recognition, each member played an important role in the final product, resulting in us creating a functional and well-integrated system.